

Classification of Movie Character Speech by Movie Genre

Claire Wang

1 Introduction

Movies are commonly labeled by genre, which can denote anything from a medium (e.g. animation) to a setting (e.g. Western). Since a movie can have more than one genre, and there is no single aspect of it that can be used to determine its genre(s), classification of movies by genre poses an interesting challenge for machine learning.

In this project, I attempted to perform this classification solely based on character dialog. I wanted to investigate whether a model could predict genre without any of the visual input that a human watching a movie would receive.

Additionally, genres often have conventions such as stock characters or common plot devices. I was interested in the question of whether these conventions are present in dialog to such an extent that a model could identify them. A system that predicts what genre(s) a character seems to be from could be of use to a screenwriter working on a script or a film studio determining how to market a movie.

2 Data

The dataset I used was the [Cornell Movie-Dialogs Corpus](#), a collection of conversations extracted from movie scripts that were retrieved from publicly accessible URLs. The genre labels and other movie metadata in the corpus were obtained from the Internet Movie Database (IMDb). The data was formatted as Conversation objects made up of Utterance objects, where each Utterance was linked to a Speaker object. The ConvoKit toolkit associated with the corpus provided methods for iterating through and retrieving metadata from these objects.

The full dataset contains 304,713 utterances from 9,035 characters from 617 movies. There are 24 distinct genre labels (Figure 1 lists them along with the number of movies each applies to). Figure 2 shows the number of genres per movie, which

was 3.0 on average. One movie (*Labor of Love*) was labeled with 0 genres, which was most likely an error. Several movies had duplicate genre labels, which was not an issue after data preprocessing.

2.1 Data Preprocessing

Each of my data points was the collection of all utterances by a single character, split into tokens by whitespace and punctuation. I removed all words from the English stopwords list provided by the Natural Language Toolkit (NLTK). I also removed names of other characters from the movie, since I did not want the model to learn which characters were in which movie and use this information to make predictions.

I discarded 4 genre labels ("adult", "documentary", "film-noir", and "short") because they had 5 or fewer movies, which seemed like an inadequate amount of data for training. I then discarded any data points that had zero tokens or zero genre labels.

Preprocessing removed data from 73 characters, leaving me with 8,962 data points. The original dataset was not split, so I split it into train/dev/test sets using a 70/15/15 ratio. This left me with 6,272 training data points, 1,345 dev data points, and 1,345 test data points.

3 Dev Set Results

To train and evaluate models, I used the scikit-learn library. I performed a grid search using 3 feature sets, 2 model types, and 3-4 hyperparameter values for each model type (except Gaussian Naive Bayes). My results are shown in Table 1.

3.1 Features

The feature sets I used were unigrams with counts, 50-dimensional word embeddings, and 100-dimensional word embeddings. For the unigrams, I converted the lists of tokens obtained

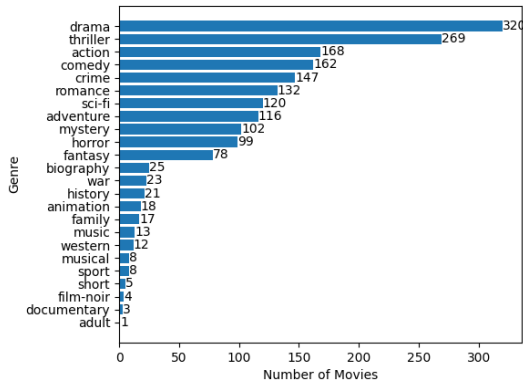


Figure 1: Frequency of genre labels in the Cornell Movie-Dialogs Corpus.

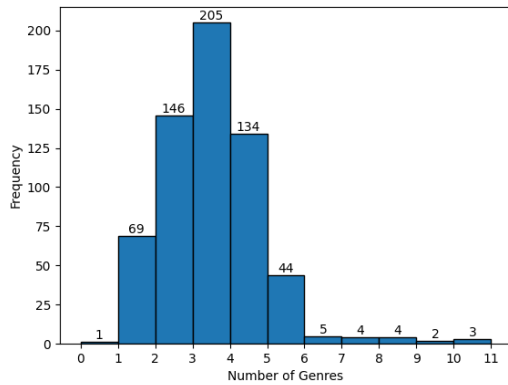


Figure 2: Number of genres per movie in the Cornell Movie-Dialogs Corpus.

from preprocessing into Counters and then used DictVectorizer to convert the Counters into vectors. Tables 4 and 5 display the 50 most frequent unigrams for each genre. I bolded ones that were unique or seemed notable.

For the 50-dimensional embeddings, I found the vector equivalent of each token from the model glove-twitter-50 and used the average of the vectors. For the 100-dimensional embeddings, I did the same with model glove-wiki-gigaword-100.

3.2 Models

I used MultiOutputClassifier, which fits one classifier per label, with both Naive Bayes and logistic regression. For unigrams, I used multinomial Naive Bayes with $\alpha = 0.05, 0.1$, and 0.2 . For embeddings, I used Gaussian Naive Bayes, which does not have an α parameter. For all feature types, I tested logistic regression with $C = 0.5, 1.0, 2.0$, and 10.0 .

I also experimented with using ClassifierChain instead of MultiOutputClassifier. ClassifierChain achieved very similar but slightly worse results for most configurations, so I have omitted them for the sake of brevity.

3.3 Metrics

I evaluated model performance using Hamming loss, weighted precision, and weighted recall. I did not use accuracy because it only considered whether the set of predicted labels for a data point exactly matched the set of true labels, and I wanted to use metrics that would take partially correct predictions into account.

4 Test Set Results

The unigrams with Naive Bayes configurations performed better than other configurations on the dev set, so I only ran these three configurations on the test set. My results are shown in Table 2.

$\alpha = 0.05$ had the best Hamming loss and precision, while $\alpha = 0.1$ had the best recall. I designated $\alpha = 0.1$ as the best performing configuration because it had a better balance between precision and recall and the difference in Hamming loss was only 0.10 . Table 3 and Figure 3 show more detailed metrics for this configuration.

4.1 Error Analysis

Since recall was notably worse than precision for most labels, I created Figure 4 to visualize the most

Configuration	HL (%)	P (%)	R (%)
Unigrams NB, $\alpha = 0.05$	9.42	70.09	60.94
Unigrams NB, $\alpha = 0.1$	9.47	69.32	61.85
Unigrams NB, $\alpha = 0.2$	9.58	68.67	61.75
Unigrams LR, $C = 0.5$	11.82	62.71	42.90
Unigrams LR, $C = 1.0$	12.14	60.35	43.85
Unigrams LR, $C = 2.0$	12.42	58.72	45.04
Unigrams LR, $C = 10.0$	13.08	55.42	46.11
Embeddings 100d NB	28.96	37.04	54.73
Embeddings 100d LR, $C = 0.5$	13.23	48.64	25.84
Embeddings 100d LR, $C = 1.0$	13.28	50.51	26.53
Embeddings 100d LR, $C = 2.0$	13.27	51.85	27.31
Embeddings 100d LR, $C = 10.0$	13.30	50.00	28.13
Embeddings 50d NB	22.96	38.71	48.55
Embeddings 50d LR, $C = 0.5$	13.39	46.82	23.78
Embeddings 50d LR, $C = 1.0$	13.44	46.34	24.11
Embeddings 50d LR, $C = 2.0$	13.46	46.93	24.57
Embeddings 50d LR, $C = 10.0$	13.48	46.90	24.80

Table 1: Model configurations used on dev set and their performance metrics: Hamming loss (HL), weighted precision (P), and weighted recall (R).

common false negatives. Each cell contains the number of data points that had the true label and were classified as the predicted label, divided by the total number of data points with the true label. For cells where the true and predicted labels are the same (on the upper left to lower right diagonal), the value is equivalent to recall. When they are not, the value is the proportion of data with the true label that was incorrectly classified as the predicted label.

5 Discussion

My model did not perform as well as I had hoped, but I think it was a reasonable first attempt for a multilabel classification task with many labels.

5.1 Data

The genre labels from my dataset were not completely accurate, as evidenced by the movie labeled with zero genres and a few mistakes I noticed when manually inspecting the data. Ideally, I would have liked to correct these errors by validating the genres against another dataset or scraping them from IMDb myself. I could also have experimented with a different set of genre labels. For instance, it may not make sense to predict whether a movie is an

animation based on its dialog.

5.2 Features

Looking at the most frequent unigrams in Tables 4 and 5 revealed some interesting patterns. Results are very similar across genres, but some genres had more unique unigrams that seemed relevant to their subject: "money" and profanity in crime, "night" and "dead" in horror, "game" and "play" in sport, "war" and "men" in war. "U" was present for many labels, most likely due to occurrences of U.S. and U.K. (in future work, I could correct this to keep acronyms as one token). Masculine terms such as "man", "mr", and "sir" appear frequently, but there are no occurrences of feminine terms.

I expected better performance from the embeddings, but the most frequent unigrams across labels were very similar, so the average of all unigram embeddings was likely too coarse of a metric to work well. Using a method besides average or excluding some of the common shared unigrams might result in better performance. The 100-dimensional embeddings outperformed the 50-dimensional embeddings for all configurations, which matched my expectations.

If I were to continue this work, I would experiment with other methods of feature extraction. I

Configuration	HL (%)	P (%)	R (%)	F1 (%)
Unigrams NB, $\alpha = 0.05$	9.98	68.49	58.30	61.95
Unigrams NB, $\alpha = 0.1$	10.08	67.32	59.37	62.25
Unigrams NB, $\alpha = 0.2$	10.39	65.52	58.79	61.17

Table 2: Model configurations used on test set and their performance metrics: Hamming loss (HL), weighted precision (P), weighted recall (R), and weighted F1.

	P (%)	R (%)	F1 (%)	Support
Drama	74.97	84.22	79.33	754
Thriller	73.59	67.92	70.64	558
Comedy	61.68	56.59	59.03	364
Action	67.30	52.52	59.00	337
Crime	61.75	66.99	64.26	306
Romance	54.77	52.36	53.54	296
Sci-Fi	73.46	52.42	61.18	227
Mystery	66.67	44.34	53.26	221
Adventure	63.07	52.36	57.22	212
Horror	73.91	46.45	57.05	183
Fantasy	61.90	35.14	44.83	148
Biography	63.83	44.78	52.63	67
War	69.05	49.15	57.43	59
History	61.90	35.14	44.83	58
Animation	44.44	11.76	18.60	34
Western	57.14	25.81	35.56	31
Family	50.00	19.35	27.91	31
Music	77.78	28.00	41.18	25
Musical	66.67	12.50	21.05	16
Sport	28.57	18.18	22.22	11
Micro Avg	67.77	59.37	63.29	3938
Macro Avg	62.38	43.46	49.41	3938
Weighted Avg	67.32	59.37	62.25	3938
Samples Avg	66.88	62.14	61.20	3938

Table 3: Per-label and average metrics for test set on best-performing configuration (unigrams with Naive Bayes, $\alpha = 0.1$). Per-label metrics are ordered by support (descending).

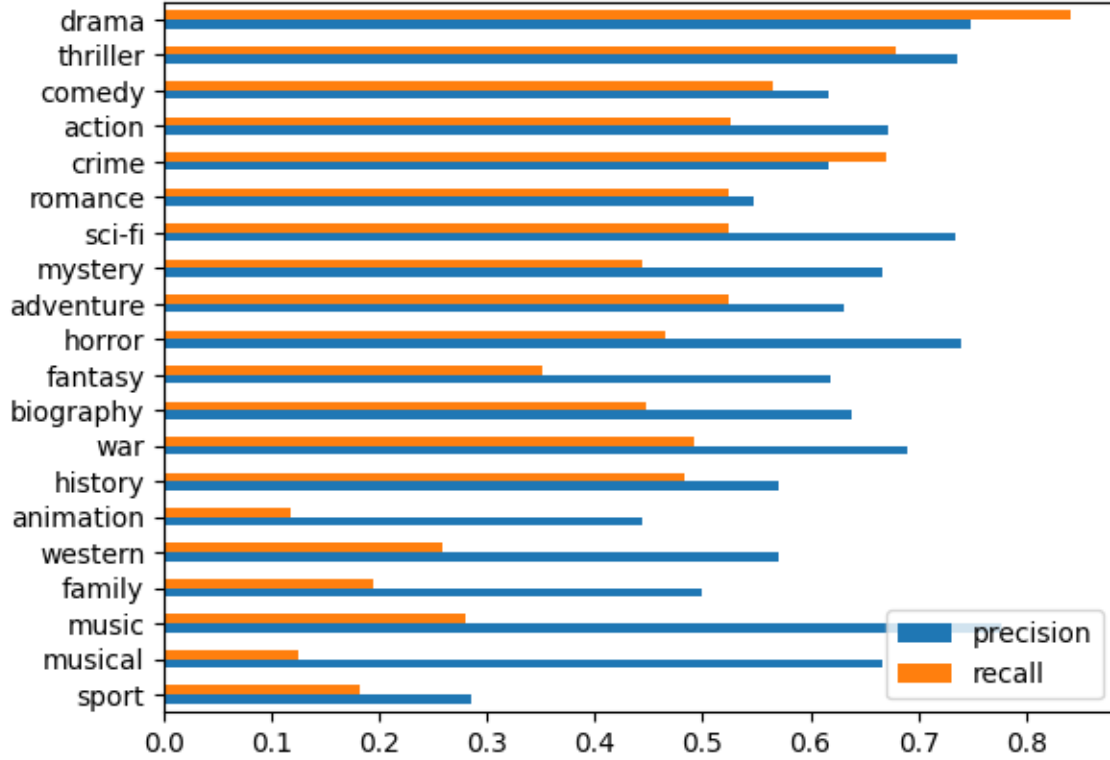


Figure 3: Per-label precision and recall for test set on best-performing configuration (unigrams with Naive Bayes, $\alpha = 0.1$). Labels are in the same order as Table 3 (from highest to lowest support).

used unigrams in an attempt to find patterns in the words characters used and the topics they were speaking about, but features such as bigrams, sentence length, punctuation, or features related to the conversation each utterance was from might yield better results. I could also expand the stop words list to remove very frequent tokens such as "know" and "like".

5.3 Models

I was somewhat surprised that Naive Bayes outperformed logistic regression for both unigrams. For embeddings, Gaussian Naive Bayes had higher Hamming loss and lower precision but higher recall than logistic regression, which I found interesting and may be worth further investigation.

5.4 Errors

As shown in Figure 3, recall was notably worse than precision for most labels and positively correlated with support. From Figure 4, we can see that the most common errors came from the model overpredicting the most frequent labels ("drama", "comedy", "thriller"). I believe these issues were caused by imbalances in the data. In future work, I could try to address this by balancing the train-

ing data, but this would significantly reduce the amount of training data, which could pose its own challenges.

6 Conclusion

In classifying movie character speech by genre, I achieved the best results using unigram features and a Naive Bayes model with $\alpha = 0.1$. If I were to continue this work, I would fix labeling errors and attempt to balance the data as well as try different methods of extracting unigram/embedding features or other feature types. I would also experiment with more model configurations (e.g. different solvers for logistic regression, other model types, alternatives to `MultiOutputClassifier`.)

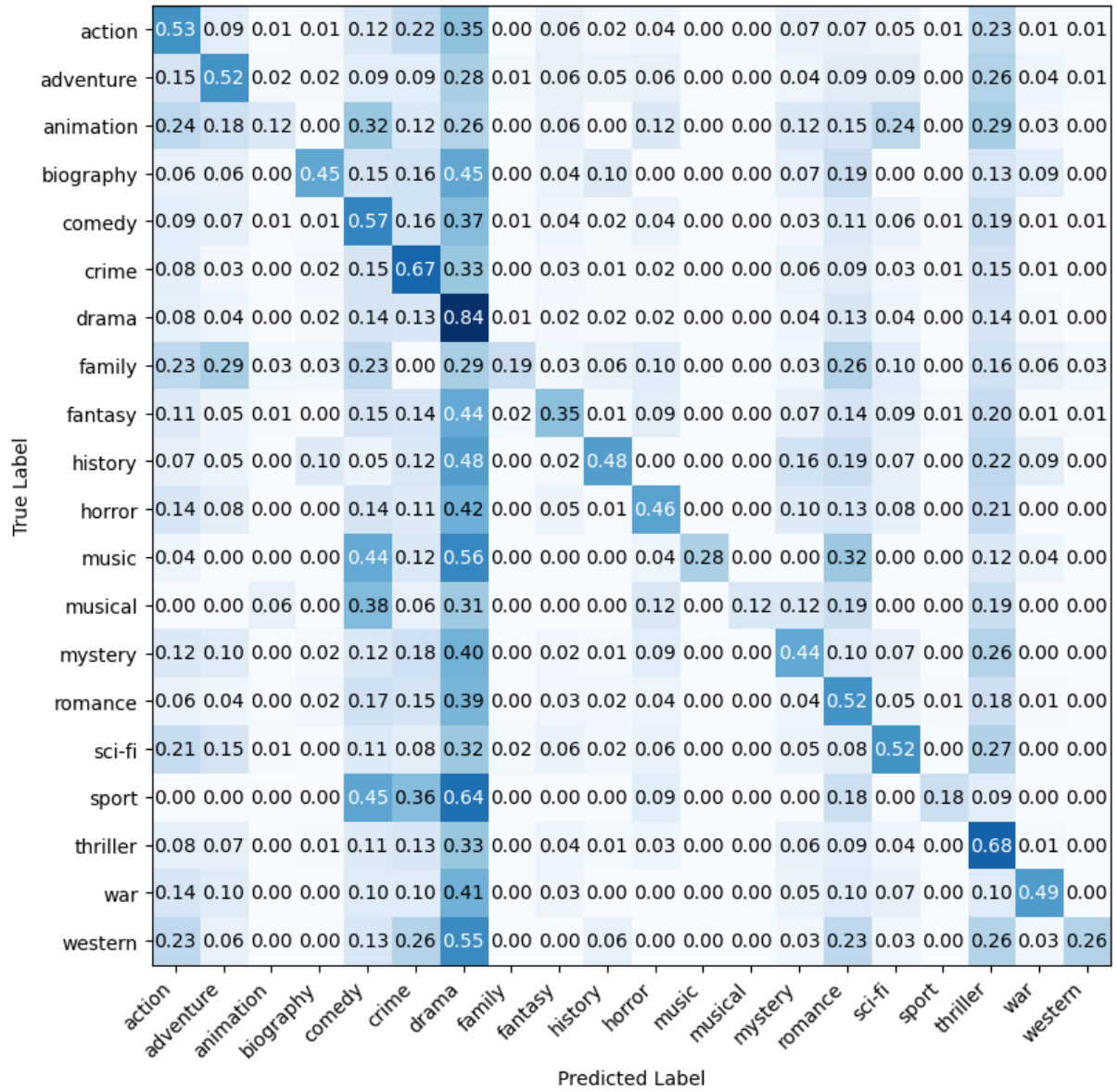


Figure 4: Matrix showing recall / false negative error rate by label. See Section 4.1 for an explanation.

Genre	Most Frequent Unigrams
Action	know, get, like, one, got, right, think, go, want, going, us, see, u, well, back, good, time, let, come, could, man, tell, way, would, take, oh, yes, look, something, yeah, never, say, make, need, two, okay, sir, little, thing, sure, gonna, mean, maybe, people, give, find, really, much, still, said
Adventure	know, get, one, like, got, right, go, going, well, think, want, oh, see, us, yes, come, u, time, good, could, back, let, take, tell, would, way, never, look, say, something, man, sir, make, need, must, two, yeah, thing, find, little, sure, people, okay, much, mean, give, mr, life , sorry, really
Animation	u , know, get, got, right, huh , well, one, like, oh, yeah, think, see, go, want, tell, going, back, time, would, come, heh , us, good, man, gonna, yes, let, could, take, something, say, way, make, uh, hey, never, look, mr, find, need, mean, thing, okay, sure, really, much, must, people, help
Biography	know, like, want, get, got, one, well, think, right, yes, good, go, oh, yeah, see, mr, tell, going, say, would, gonna, take, look, man, people, never, time, could, let, back, come, u, us, make, mean, something, way, much, need, talk, really, sure, thing, money , said, give, work , even, ever, nothing
Comedy	know, like, get, well, oh, u, think, got, want, right, one, go, going, yeah, see, good, yes, say, look, let, tell, time, come, really, would, back, okay, mean, could, something, never, take, us, man, little, make, gonna, way, sure, thing, people, love , maybe, said, sorry, two, need, hey, uh, give
Crime	know, get, like, got, want, one, think, right, go, yeah, well, see, going, tell, good, u, let, back, us, gonna, time, take, would, come, look, could, say, yes, oh, man, never, way, something, money , mean, okay, make, sure, need, little, two, shit , thing, said, maybe, give, guy , fuck , much, really
Drama	know, like, get, want, got, think, well, one, go, right, going, see, good, yes, oh, yeah, tell, come, would, say, time, back, look, let, take, could, never, us, man, mean, something, way, u, make, really, gonna, okay, sure, people, little, said, mr, maybe, thing, need, much, two, give, talk, anything
Family	oh , know, well, get, like, right, think, one, see, yes, got, go, come, u, look, good, would, going, want, back, time, way, never, could, say, yeah, let, okay, tell, mean, gonna, help, really, people, us, sure, take, something, thing, make, great, said, little, nothing, maybe, hey, give, home , please, much
Fantasy	know, like, get, u, one, oh, got, think, want, well, right, going, go, see, come, yes, good, time, could, us, never, back, let, would, look, tell, take, something, say, way, little, mean, make, man, need, yeah, sure, mr, must, thing, okay, really, people, help, much, maybe, two, give, said, nothing
History	know, one, like, get, got, think, go, yes, right, want, well, mr, us, going, would, man, good, never, say, back, see, let, could, come, sir, time, tell, must, people, make, look, oh, way, take, sure, even, much, need, god , nothing, something, ever, day, talk, said, name, please, john , thing, anything

Table 4: 50 most frequent unigrams per genre in training data. (1/2)

Genre	50 Most Frequent Unigrams
Horror	know, get, like, think, got, right, one, go, want, going, well, see, back, us, oh, yeah, something, come, good, time, let, u, look, yes, could, tell, way, would, okay, take, sure, maybe, really, never, say, mean, little, make, thing, anything, need, man, gonna, said, night , nothing, god , dead , two, last
Music	know, well, like, one, got, oh, get, want, right, go, see, think, yes, yeah, good, let, time, come, us, look, would, really, going, mean, say, little, okay, never, could, give, tell, said, thing, gonna, something, make, two, people, take, back, show , man, way, need, sure, much, love , even, maybe, old
Musical	oh , get , know, like, well, see, got, yes, come, right, think, going, go, want, say, one, u, back, let, good, time, us, could, help, take, would, little, way, something, never, man, make, must, sure, look, tell, yeah, away, b , home , give, dude , people, la , heart , dear , need, ever, war , place
Mystery	know, like, get, one, got, think, want, going, right, go, well, see, tell, u, would, yes, time, something, back, come, say, could, good, us, oh, yeah, look, let, never, way, take, mean, mr, said, okay, make, anything, maybe, people, man, need, little, sure, thing, two, even, really, gonna, call , told
Romance	know, like, get, well, want, think, right, got, one, go, oh, going, yes, see, good, yeah, say, tell, come, time, would, look, back, could, never, let, take, really, mean, something, love , okay, little, u, way, make, sure, us, man, said, maybe, thing, gonna, much, people, two, give, sorry, thought, mr
Sci-Fi	know, get, like, one, think, right, go, got, going, want, see, well, us, time, could, yes, come, back, something, good, would, u, oh, way, sir, let, tell, take, look, never, need, make, say, mean, man, yeah, thing, little, maybe, sure, okay, two, people, really, find, anything, still, said, even, must
Sport	know, ya , like, get, got, want, right, go, gonna, think, yeah, say, good, see, one, man, tell, let, make, look, well, oh, never, time, love , gotta , take, way, hey, back, ever, talk, game , said, mean, sure, come, okay, money , going, something, bad , maybe, cause, two, little, play , around, give, could
Thriller	know, get, like, got, think, one, want, right, go, going, well, u, see, tell, good, time, back, us, could, come, would, yeah, yes, let, look, something, take, oh, say, way, never, okay, man, sure, make, need, mean, gonna, said, two, maybe, little, thing, people, really, mr, anything, call , much, give
War	know, get, one, like, us, yes, go, think, got, want, good, going, see, well, would, right, man, come, take, back, let, sir, never, time, must, could, say, tell, make, much, oh, way, said, god , people, please, war , need, men , look, love , give, mean, yeah, find, little, two, even, always, nothing
Western	know, got, like, get, well, right, one, go, want, back, think, good, going, come, see, time, way, say, would, yeah, tell, us, oh, could, look, never, em , man, maybe, people, take, gonna, sir, mean, little, let, said, make, sure, yes, two, around, much, place , need, thing, thought, even, money , ever

Table 5: 50 most frequent unigrams per genre in training data. (2/2)